

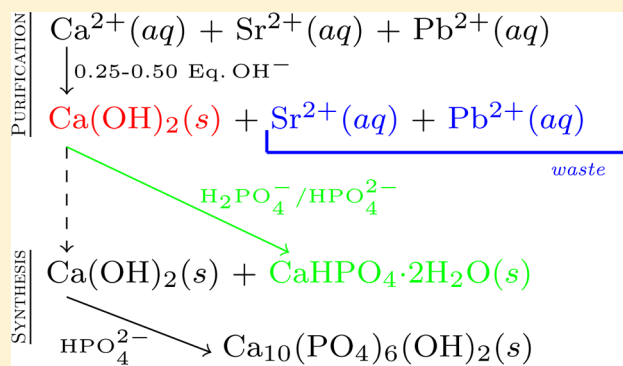
Pure Hydroxyapatite Phantoms for the Calibration of in Vivo X-ray Fluorescence Systems of Bone Lead and Strontium Quantification

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ABSTRACT: Plaster of Paris [poP, $\text{CaSO}_4 \cdot (1)/(2) \text{H}_2\text{O}$] is the standard phantom material used for the calibration of in vivo X-ray fluorescence (IVXRF)-based systems of bone metal quantification (i.e. bone strontium and lead). Calibration of IVXRF systems of bone metal quantification employs the use of a coherent normalization procedure which requires the application of a coherent correction factor (CCF) to the data, calculated as the ratio of the relativistic form factors of the phantom material and bone mineral. Various issues have been raised as to the suitability of poP for the calibration of IVXRF systems of bone metal quantification which include its chemical purity and its chemical difference from bone mineral (a calcium phosphate). This work describes the preparation of a chemically pure hydroxyapatite phantom material, of known composition and stoichiometry, proposed for the purpose of calibrating IVXRF systems of bone strontium and lead quantification as a replacement for poP. The issue with contamination by the analyte was resolved by preparing pure $\text{Ca}(\text{OH})_2$ by hydroxide precipitation, which was found to bring strontium and lead levels to <0.7 and $<0.3 \mu\text{g/g}$ Ca, respectively. HAP phantoms were prepared from known quantities of chemically pure $\text{Ca}(\text{OH})_2$, $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ prepared from pure $\text{Ca}(\text{OH})_2$, the analyte, and a HPO_4^{2-} containing setting solution. The final crystal structure of the material was found to be similar to that of the bone mineral component of NIST SRM 1486 (bone meal), as determined by powder X-ray diffraction spectrometry.



Human bone is used as a biomarker of cumulative exposure to various elements including lead and strontium. In vivo X-ray fluorescence spectrometry (IVXRF) is an analytical methodology which allows one to quantify the concentration of such elements in bone nondestructively and with minimal radiation dose to the subject. IVXRF-based bone metal quantification has traditionally been complicated by the complex nature of the analyte's characteristic X-ray intensity which influences the calibration protocol. By extension, a major hindrance to the calibration of IVXRF systems of bone metal quantification is the availability of suitable calibrators, otherwise and hereby referred to as phantoms, to be used for the purpose of system calibration.

The earliest IVXRF-based system of bone metal quantification is the ^{57}Co -induced K-IVXRF system described by Ahlgren et al.¹ and Ahlgren and Mattsson,² designed for the purpose of determining absolute bone lead concentrations in the finger bone of human subjects. Calibration and reliable concentration extraction from the IVXRF spectrum were hindered by various factors, including intersubject variations in bone geometry and bone mineral concentration (i.e., gross bone composition).² Calibration thus required that the bone size of the subject be determined by the acquisition of a series of orthoplanar X-rays of the measurement site and phantoms developed from bone ash and silica wax to mimic both the bone size and the bone

mineral concentration of the subject's bone, estimated from the scattered source radiation.² In subsequently investigated IVXRF-based methods of bone lead and strontium quantification, because of such challenges, it was necessary to resort either to the complete absence of a calibration procedure or to a calibration procedure based on the measurement of bone specimens which had been cross validated for their bone metal concentrations by a secondary method of analysis.^{3–7} Such an approach resulted in limited concentration ranges included in the calibration and lacked a true blank calibrator.^{3–7}

A major improvement to the K-IVXRF analysis of bone lead was the introduction of a ^{109}Cd source as a replacement for the ^{57}Co source.⁸ After this major development, Somervaille et al.⁹ introduced a normalization procedure to the calibration protocol of the ^{109}Cd -induced K-IVXRF system of bone lead quantification which has become standard to IVXRF bone lead analysis and by extension to other IVXRF bone metal analyses. It was found that the majority of the coherently scattered ^{109}Cd γ -rays had origins from scattering events within the bone mineral.^{9–11} By normalizing the characteristic X-ray intensity to

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that of the coherently scattered source radiation intensity, it became possible to determine the quantity of lead per unit mass of bone mineral. This normalization procedure corrected the observed response for various factors, including variations in source activity, source-to-detector distance, soft tissue thickness and associated signal attenuation, variations in bone shape, size, and orientation, subject bone mineral concentration, and minor subject movement during measurement.^{9–11}

The fact that coherently scattered ^{109}Cd γ -rays originate predominately in bone mineral allows for the introduction of calibrators consisting of plaster of Paris [poP, $\text{CaSO}_4 \cdot (1)/(2)\text{H}_2\text{O}$] doped with the analyte. Concentrations are extracted from an in vivo measurement by extrapolation after applying a pseudomatrix correction to account for the different scattering properties between the phantom (predominately a calcium sulfate) and the bone mineral being measured (predominately a calcium phosphate). The matrix correction factor is simply the ratio of the relativistic form factors of both poP and bone mineral, assumed to be pure hydroxyapatite [HAp, $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$]; the correction factor itself is referred to as the coherent conversion factor (CCF).^{9,11} The success of coherent normalization-based calibration in the context of bone lead quantification by ^{109}Cd -induced K-IVXRF^{9,12–17} has resulted in it becoming the most utilized normalization procedure within the calibration protocol of IVXRF-based systems for quantifying bone uranium,^{18,19} strontium,^{20–24} lead via L-IVXRF^{25–28} and ^{57}Co -induced K-IVXRF.²⁹

The use of poP as the phantom material raises some concerns about its purity and chemical composition in comparison to the matrix being measured. In the case of bone lead analysis, contamination in the poP used for the preparation of phantoms has been insinuated as a problem due to assumptions of the purity of the material when computing the CCF.^{11,30} It has been proposed that the level of contamination may influence the reliability of the applied CCF given that poP used for phantom preparation is generally not pure poP and does present some contamination, which is not necessarily accounted for.^{11,30}

The problem with contamination in poP is much more severe in the case of in vivo bone strontium analysis.^{20,30} Strontium, being a ubiquitous element to calcium, is generally present in calcium compounds at levels which are near those of the levels expected in human bone. This is a feature which is independent of the quoted degree of purity of calcium compounds.^{20,31} The fact that strontium is present in poP at high concentrations does not allow for the development of a proper calibration procedure. Strontium contamination does not allow for the preparation of a blank phantom, making the true analytical figures of merit of the system difficult to determine.³¹

The chemical difference between the poP, a calcium sulfate (albeit also containing other species of calcium^{11,30}), and bone mineral, a calcium phosphate, also presents some general concerns. The application of a CCF inherently introduces some uncertainty in the quantification while the chemical difference between the two materials, in the context of ^{109}Cd induced K-IVXRF bone lead measurements, results in different high-energy tails from the Compton scattering profile for calcium, sulfur, and phosphorus.^{32–35} As such, having a phantom material composed of a calcium sulfate and a sample of calcium phosphate results in different spectral features, which require different data analysis procedures. It is desirable to have a phantom material that mimics bone mineral (HAp) as closely

as possible while also being chemically pure. This work describes a simple method of preparing pure hydroxyapatite (HAp) phantoms as a more suitable phantom material for bone strontium and lead quantification using IVXRF.

■ EXPERIMENTAL SECTION

Preparation of Pure Calcium Compounds. All of the reagents used for the preparation of phantoms and/or the purification of calcium were assessed for purity by total reflection X-ray fluorescence spectrometry (TXRF). These reagents included calcium chloride dihydrate ($\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$, 99.0%+, Sigma-Aldrich, St. Louis, MO), calcium hydrogen phosphate dihydrate ($\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$, USP grade, Amresco, Solon, OH), calcium hydroxide [$\text{Ca}(\text{OH})_2$, USP grade, Amresco, Solon, OH, USA], sodium hydroxide (NaOH, ACS grade, BMD, West Chester, PA), sodium hydrogen phosphate anhydrous (Na_2HPO_4 , ACS grade, Amresco, Solon, OH), sodium phosphate monobasic monohydrate ($\text{NaH}_2\text{PO}_4 \cdot \text{H}_2\text{O}$, ACS grade, BMD, West Chester, PA), metal grade HNO_3 (69.0–70.0%; EMD Millipore, VWR Scientific, Mississauga, Ontario, Canada), and 18.2 M Ω cm water drawn from a Milli-Q system.

Low strontium $\text{Ca}(\text{OH})_2$ was prepared via a method similar to that described by Patti and Hernandez.³⁶ A 2 mol/dm³ aqueous solution of Ca^{2+} was prepared from $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$. An aqueous solution of NaOH, made to the appropriate concentration, was then added to the Ca^{2+} solution to precipitate $\text{Ca}(\text{OH})_2$. The solutions were mixed such that the final total calcium concentration in the solution was 1 mol/dm³. The precipitate was allowed to age in the mother liquor for a period of 2 h prior to recovery by vacuum-assisted filtration over an ashless Watman filter paper (#512) with several washings of water. The $\text{Ca}(\text{OH})_2$ was dried at 110 °C in a conventional gravity drying oven for a period of 6 h. The dried $\text{Ca}(\text{OH})_2$ was milled to a fine powder in a tungsten carbide ball mill at a frequency of 30 Hz for a total milling time of 20 min prior to use. The $\text{Ca}(\text{OH})_2$ was assessed for purity by TXRF and energy dispersive X-ray fluorescence spectrometry (EDXRF) and the crystal phase confirmed by powder X-ray diffraction spectrometry (XRD).

Low strontium $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ was prepared from the low strontium $\text{Ca}(\text{OH})_2$. A slurry was prepared from a suitable mass of $\text{Ca}(\text{OH})_2$ in water. The $\text{Ca}(\text{OH})_2$ was dissolved by the slow dropwise addition of a 1 mol/dm³ aqueous solution of HNO_3 . The aqueous solution was brought to a Ca^{2+} concentration of 0.14 mol/dm³ and adjusted to a pH of 6.9 by the dropwise addition of a 0.05 mol/dm³ aqueous solution of NaOH (if required). A second aqueous solution containing 0.030 mol/dm³ Na_2HPO_4 and 0.008 mol/dm³ $\text{NaH}_2\text{PO}_4 \cdot \text{H}_2\text{O}$ was prepared. The solutions were prepared such that the mole ratio between the calcium ion and both phosphoric acid anions combined was 1:1. The solution containing the phosphoric acid anions was quickly added to the calcium-containing solution, and the solution was stirred vigorously for a period of 2 h. The $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ was then recovered by vacuum-assisted filtration over an ashless Watman filter paper (#512) with several washings of water. The $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ was dried at 110 °C for a period of 6 h prior to characterization by TXRF and XRD.

Preparation of Hydroxyapatite Phantoms. Phantoms were prepared by mixing a suitable amount of dry $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ and $\text{Ca}(\text{OH})_2$ in order to achieve a 1.67 Ca/P mole ratio; the Ca/P mole ratio of HAp. The powders were weighed

to the nearest 0.01 mg on an analytical balance calibrated against a 200.0000 g internal calibration mass (GR-202, A&D Company Ltd., Tokyo, Japan). The powders were dry mixed by rotation in an acid-washed polypropylene container in which the powders did not occupy more than 10% of the total volume. The phantoms were set by adding them to a 1 mol/dm³ solution of NaHPO₄ at a powder-to-liquid mass ratio of 2:1, followed by thorough mixing for a period of ca. 1 min and by transfer of the paste to suitable molds. The molds used for the preparation of phantoms in this study consisted of Mylar XRF sample cups with a 10 mm diameter aperture and a height of 25.1 mm (Chemplex Industries, Palm City, FL) sealed at one end with 4 μ m thick Prolene (Chemplex Industries, Palm City, FL). The phantoms were allowed to set for 24 h at 100% humidity, were unmolded, and were allowed to cure for an additional 24 h at 100% humidity. The phantoms were set under ambient conditions for 7 days until they reached constant weight. After a further 7 days stored under ambient conditions, the phantoms were dried in a drying oven at 105 °C for 24 h, resulting in the final phantom. The crystal phase of the phantom material was assessed by XRD and compared to the crystal phase of bone mineral against NIST bone meal (SRM 1486; National Institute of Standard and Technology, Gaithersburg, MD). The density of the phantom material was determined via Archimedes' method in water.

The addition of the analyte was performed gravimetrically by adding, to the nearest 0.01 mg, a suitable amount of a NIST traceable 1000 ppm inductively coupled plasma atomic emission spectrometry standard solution of the analyte of interest (ULTRAScientific, N. Kingstown, RI). The analyte solution was added to the 1 mol/dm³ NaHPO₄ setting solution, which resulted in the formation of a fine and well-dispersed precipitate. The powders were added immediately to the analyte-containing setting solution and the phantom prepared as described previously. Standard curves were prepared and measured by EDXRF.

Total Reflection X-ray Fluorescence Spectrometry. High-purity quartz reflectors were used as the sample carriers (Bruker-AXS, Madison, WI). The reflectors were cleaned by placing them into a 2% RBS 50 (Sigma-Aldrich, Oakville, Ontario, Canada) solution prepared with Milli-Q water and microwaving at full power for a period of 10 min. The reflectors were removed from the 2% RBS 50 solution and thoroughly rinsed with water. The reflectors were rinsed in acetone, allowed to dry, and the RBS 50 wash repeated with a 1% RBS 50 solution followed by rinsing in water. The reflectors were then treated, for a period of 3 h, in a 10% HNO₃ solution. The HNO₃ bath was maintained at 80 °C using a conventional hot plate during this cleaning phase. After rinsing with water, the reflectors were microwaved for a further 10 min in water and dried in a drying oven at 85 °C for 45 min. All reflectors were checked for contamination at this stage by collecting a spectrum for 3600 s live time.

TXRF measurements were performed on an S2 PicoFox spectrometer (Bruker-AXS). The source photons consisted of monochromatic 17.5 keV X-rays produced from a molybdenum target X-ray tube operating at 50 kV and 600 μ A. The X-ray tube was focused such that the spot size on the sample was approximately (50 $\times$ 50) μ m². Photon detection was performed using a silicon drift detector (SDD) with a total area of 30 mm² positioned approximately 2 mm from the sample carrier surface. Signal processing was performed on a 4096 channel 12 bit analogue-to-digital converter (ADC). The detection system had

an energy resolution of 155 eV, as measured through the Mn K α line at ca. 10 kcps. Prior to all analyses, the detector gain was checked and adjusted using a 1 μ g arsenic standard (Bruker-AXS). The sensitivity of the system for all elements was determined using a 1 ng Ni standard (Bruker-AXS). The sensitivity for Ni was then used as a means of calculating the theoretical sensitivities for all elements and for the relevant X-ray series. These sensitivity factors were used for the calculation of the total mass of a given element present within the spot size of the instrument. The quantification method was checked using a 1 μ L deposition of a 1 μ g/mL multielemental standard (Bernd Kraft GmbH, Germany).

Quantification was performed by digesting ca. 50 mg of sample in 250 μ L of HNO₃ for a period of 1 h. The solutions were brought to a volume of 1 mL with water and 1 μ L of the digest deposited and dried onto the surface of the quartz reflector. Measurements were performed for a period of 3600 s live time. Quantification was performed relative to the mass of calcium in the compounds and was performed with the accompanying SPECTRA software package. The spectra were fit using a Bayes fitting procedure and the relative quantification performed using the sensitivity factors calculated from the Ni standard. The method was validated against NIST bone ash (SRM 1400; National Institute of Standard and Technology).

Energy Dispersive X-ray Fluorescence Spectrometry. All EDXRF measurements were performed on an S2 Ranger spectrometer equipped with a Pd anode X-ray tube and an SDD detection system positioned with an incident angle of 45° and a takeoff angle of 45° (Bruker-AXS). The X-ray tube was operated with a potential of 40 kV and current of 250 μ A and the distribution hardened using a 500 μ m Al filter. Measurements were made for 1200 s live time. Total integrated peak areas of the characteristic X-rays of interest were determined using an in-house trust-region reflective least-squares curve-fitting algorithm.

Powder X-ray Diffraction Spectrometry. The crystal phase of the prepared Ca(OH)₂, CaHPO₄·2H₂O, phantoms, and NIST bone meal were assessed by powder X-ray diffraction spectrometry. The materials were finely ground prior to analysis in a tungsten carbide ball mill. The measurements were made using a Co X-ray source (λ = 1.79021 Å) operating at a potential of 40 kV and a current of 40 mA (Rigaku Geigerflex, Danvers, MA). The diffractograms were collected over the range of 5.00 to 90.00° 2 θ at a sampling interval of 0.05° and scan speed of 4°/min.

■ RESULTS AND DISCUSSION

Commercially available calcium compounds used for preparing phantoms were assessed for purity by TXRF after the method was validated against a suitable standard reference material (Table 1). All of the calcium compounds showed contamination with various transition metals as well as lead, strontium, and in the case of the calcium chloride dihydrate, bromine (Table 2).

One of the major problems associated with the use of poP as the phantom material for IXRF-based methods of bone metal quantification is that of purity. Two major sources of contamination have been identified: contamination of the whole material by the analyte and other metals and contamination of the material with other chemical species of calcium.^{11,20,30} Such contamination makes it difficult to accurately calculate the CCF or to properly determine the analytical figures of merit of IXRF systems given the inability

Table 1. Validation of the TXRF Method Used for the Quantification of Contaminants in Calcium Compounds Using NIST Bone Ash (SRM 1400)^a

	certified		determined	recovery (%)	<i>p</i>
	μg/g	μg/g Ca	μg/g Ca		
Fe	660 ± 27	1729 ± 71	1601 ± 335	93	0.32
Zn	181 ± 3	474 ± 8	630 ± 43	133	0.04
Sr	249 ± 7	652 ± 18	695 ± 32	107	0.20
Pb	9.07 ± 0.12	23.8 ± 0.3	26 ± 1	111	0.11

^aConcentrations are quoted relative to the calcium content in the bone ash of $38.18 \pm 0.13\%$ w/w. The uncertainty associated with the certified values relative to the calcium content are propagated uncertainties. Determined concentrations (by TXRF) are quoted as the mean concentration $\pm 1s$ ($n = 3$).

Table 2. Contamination Levels in the Calcium Compounds Purchased for the Preparation of Phantoms^a

	commercial compounds			purified ^b
	Ca(OH) ₂	CaHPO ₄ ·2H ₂ O	CaCl ₂ ·2H ₂ O	Ca(OH) ₂ ^c /CaHPO ₄ ·2H ₂ O ^d
Ti	63 ± 5	63 ± 8	—	—
V	—	11 ± 5	—	—
Cr	40 ± 4	30 ± 3	—	—
Mn	52 ± 3	37 ± 4	—	—
Fe	741 ± 79	1230 ± 27	75 ± 11	—
Cu	44 ± 4	—	—	—
Zn	35 ± 2	51 ± 7	24 ± 6	—
Br	—	—	480 ± 134	—
Sr	442 ± 20	286 ± 6	773 ± 241	<0.7
Pb	6 ± 1	9.8 ± 0.3	21 ± 6	<0.3

^aIn all cases, calcium compounds were contaminated with both strontium and lead, the analytes of interest, as well as various transition metals and halides. Elements either not listed or accompanied by (—) indicate that the element was present at below the limit of detection of the spectrometer. Limits of detection are listed for strontium and lead.

^bTXRF limits of detection (3σ). ^cPrepared from the CaCl₂·2H₂O.

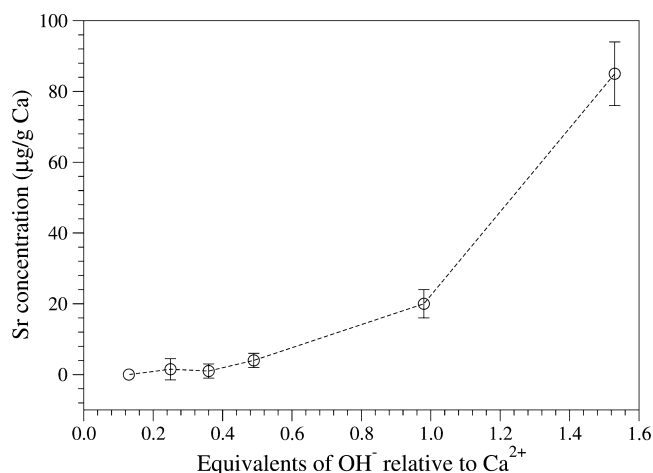
^dPrepared from the purified Ca(OH)₂.

to produce a true, well-characterized blank phantom.^{11,30,31} The contamination of calcium compounds with strontium is far more severe than contamination from lead (Table 2).^{20,30} Pure calcium compounds are required in order to have complete control over the composition of the proposed HAp phantoms.

The purification of calcium from strontium is a known challenge given the chemical similarity between the two elements. In the case in which the concentration of calcium is far greater than that of strontium, the calcium must be removed from solution, given the difficulty in removing very small masses of strontium reliably. The selective precipitation of calcium from a strontium-containing solution with 8-hydroxyquinoline has been shown to be quite successful at greatly reducing the strontium content of calcium compounds.^{37,38} However, the process is time-consuming, expensive, and requires the use of large quantities of solvent and acid for the preparation of relatively small amounts of pure calcium. That method of purification is not particularly well-suited to the preparation of a set of phantoms that require large masses of calcium compounds.

The separation of calcium from strontium has been shown to be possible by hydroxide precipitation of calcium from a strontium-containing solution.^{36,39,40} This method of separation is successful due to the differential solubility of calcium and

strontium hydroxide.³⁶ The method is quite simple, requiring only that an aqueous Ca²⁺ solution be treated with OH[−] to precipitate Ca(OH)₂. The separation of the two elements by this route is sensitive to the number of equivalents of OH[−] used for the precipitation, with a higher efficiency of purification being achieved by using only a small number of equivalents of OH[−] resulting in an excess of free Ca²⁺ in the solution (Figure 1). This observation may be due to the higher probability of

**Figure 1.** The efficiency of calcium hydroxide precipitation within a calcium chloride solution as a method for the separation of strontium from calcium as a function of the number of equivalents of OH[−] relative to Ca²⁺ used for the precipitation reaction.

coprecipitation of strontium when higher numbers of equivalents are used. In order to maximize the yield of high purity calcium, it was found that when using a calcium concentration of 1 mol/dm³, the addition of 0.5 equivalents of OH[−] produces Ca(OH)₂ with a strontium concentration of <0.7 μg/g Ca ($p = 0.05$) (Table 2), which is below the limit of detection of the current IVXRF system of bone strontium quantification.^{20,21,31} A striking feature of this method of separation is the high efficiency for not only the removal of strontium but all other elements which were originally present in the CaCl₂·2H₂O (Table 2), thus producing Ca(OH)₂ of high purity (Figure 2).

The Ca(OH)₂ produced by this method was found to be free of all metals and was used for the purpose of preparing phantoms after using the compound to prepare metal-free CaHPO₄·2H₂O (Figure 3).

The preparation of phantoms for IVXRF calibration requires that the phantoms be prepared in various shapes and sizes. The use of poP for the purpose of phantom preparation was thus ideal as, being a plaster, it can be molded and shaped to virtually any geometry. The preparation of HAp calibrators was thus accomplished by preparing an HAp cement which can similarly be molded into a phantom with the same geometric flexibility. The issue of contamination by other unknown chemical species of calcium was resolved by the preparation of the calcium salts de novo, where no sign of contamination was seen from other calcium species in the prepared compounds (Figure 3). The cement was prepared by mixing suitable quantities of CaHPO₄·2H₂O and Ca(OH)₂, as to achieve a Ca/P mole ratio of 1.67 which is that of HAp. By the addition of a setting solution which contains the HPO₄^{2−} ion, the following reaction proceeds:⁴¹

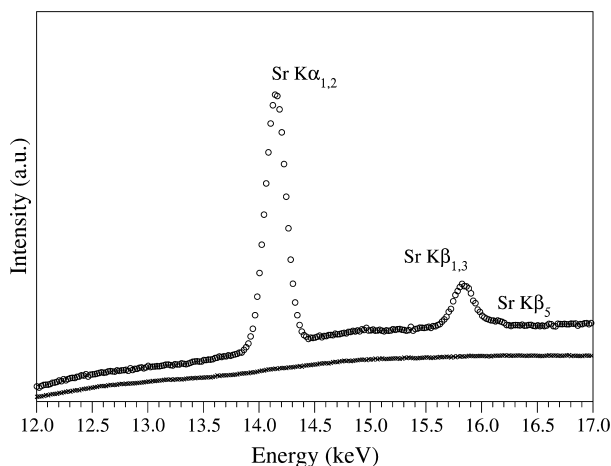


Figure 2. An EDXRF spectrum of commercially available Ca(OH)_2 (upper spectrum) and the purified Ca(OH)_2 (lower spectrum). The purified Ca(OH)_2 shows strontium levels below the limit of detection (3σ) of the EDXRF system.

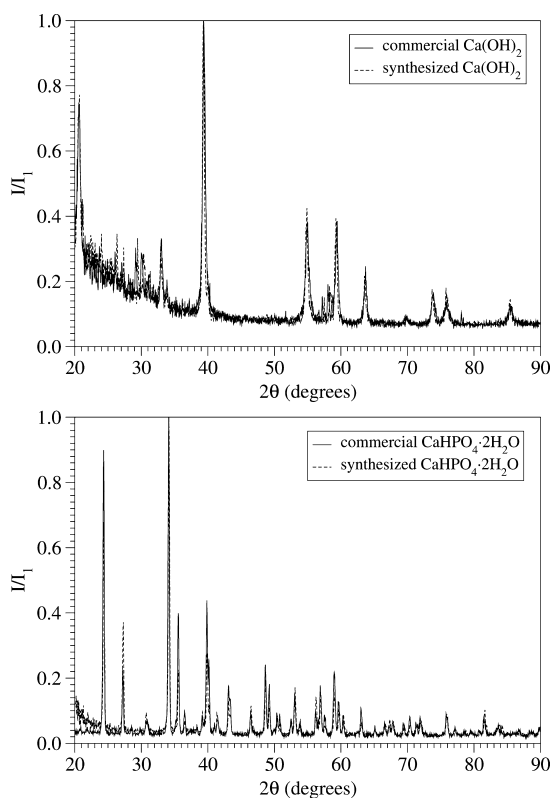
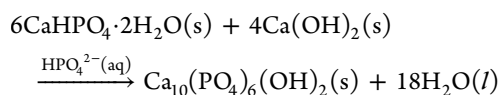


Figure 3. X-ray diffractograms of the prepared Ca(OH)_2 and $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ in comparison to commercially available materials.



producing a solid phantom composed of HAp and water. The phantoms did require setting in a humid environment, as to avoid cracking, and required on the order of 7 days to reach a constant mass once unmolded and stored at ambient. Drying of the phantom, after the complete curing phase, revealed no greater than 3% w/w water content, and the final phantoms had densities of $(2.0 \pm 0.6) \text{ g/cm}^3$. The crystal structure of the

phantom showed similarity to that of bone mineral, although a fraction of unreacted $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ remained (Figure 4). The

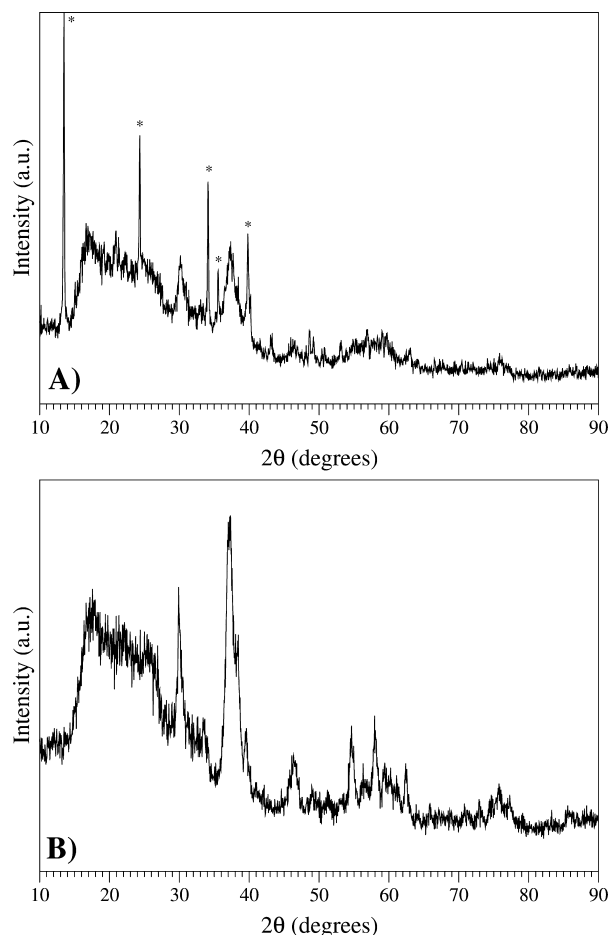


Figure 4. X-ray diffractograms of (A) HAp phantom and (B) NIST bone meal. The phantom material shows a crystal structure, indicating the production of HAp similar to bone mineral. The * indicates reflections from unreacted $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$.

fraction of unreacted $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ can be determined by mass difference given that, at least in the case of the phantoms prepared here, Ca(OH)_2 was always the limiting reagent.

The analyte was integrated into the phantom material by its addition to the setting solution. Given that both analytes are introduced as the divalent ion, the addition of the analyte to the setting solution does result in the formation of a fine precipitate of the analytes' phosphate, which presumably results in integration into the HAp lattice by substitution for $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ in the overall reaction. The integration of the analyte allowed for the preparation of standard curves, which showed the expected linearity and intercepts equivalent to 0 (Figure 5). The material thus acts as a suitable calibration material in the context of XRF analysis and given its moldable nature by extension to IVXRF applications, allowing for the production of a true blank as well as an overall chemical composition equivalent to HAp. Although the focus of this work was the development of a calibration material for IVXRF analysis, the proposed calibration material can potentially be extended as a calibrator to other analytical methodologies which currently lack sufficient calibrators.⁴²

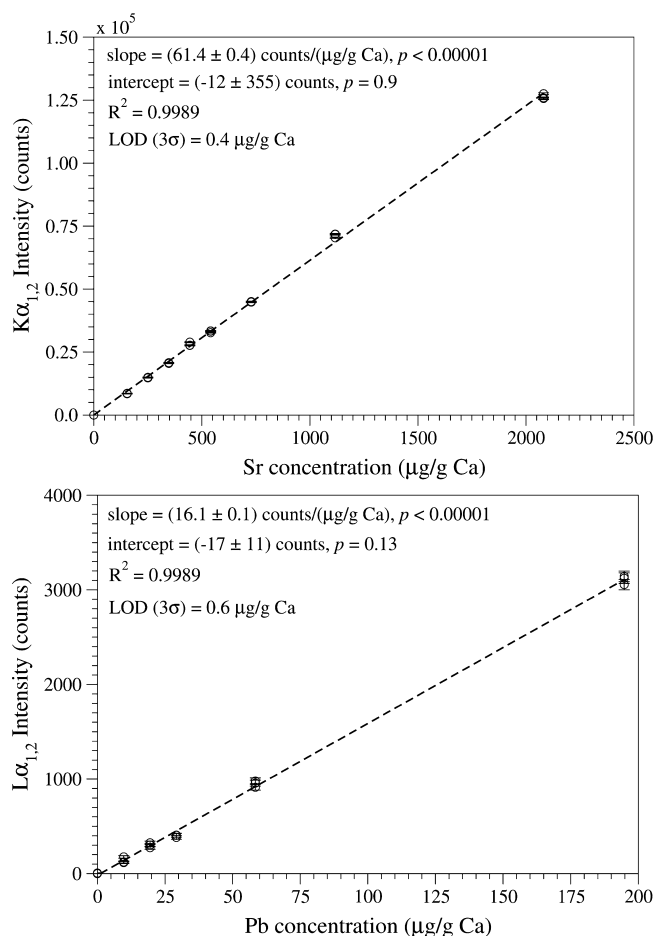


Figure 5. Calibration curves for strontium and lead prepared using HAp phantoms doped with the respective elements.

CONCLUSIONS

This work describes the preparation of a pure HAp phantom material proposed here as a material for the purpose of calibrating IVXRF systems of bone strontium and lead quantification as a replacement for poP. The issue with contamination by the analyte was resolved by preparing pure $\text{Ca}(\text{OH})_2$ by hydroxide precipitation, which was found to bring strontium and lead levels to <0.7 and $<0.3 \mu\text{g/g Ca}$, respectively. The pure $\text{Ca}(\text{OH})_2$ was then used for the preparation of analyte free $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ for the purpose of producing HAp. HAp phantoms were then prepared from the $\text{Ca}(\text{OH})_2$ and $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$ with the addition of a HPO_4^{2-} containing setting solution to which the analyte was added. The final crystal structure of the material was found to be similar to that of bone mineral (NIST SRM 1486). The phantom materials described in this work were free of the analyte and are of known composition and thus proposed as a replacement for poP in the context of IVXRF calibration. The purification protocol was found to free the calcium of all metals in comparison to the metal content found in the reagent. This may allow for the use of this phantom material as calibrators for other elements for both in vivo and ex vivo XRF applications and for general calcified tissue analysis.

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Notes

The authors declare no competing financial interest.

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